IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – [Fitness Buddy]

Domain of Project – [Healthcare / Health & Wellness / AI**]**

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
Kavita Chauhan		tavikrana@gmail.com	7081803560

Proposed Solution -

Proposed Solution: Fitness Buddy

The proposed solution is an **AI-powered Fitness Buddy** built using **IBM watsonx.ai, Granite Models, and LangGraph** to provide personalized fitness, workout, nutrition, and wellness guidance. The system addresses common challenges in maintaining a healthy lifestyle through an intelligent, conversational, and modular architecture.

At its core, the solution uses **Retrieval-Augmented Generation (RAG)** to retrieve relevant fitness and nutritional information from reliable knowledge sources. This enables the AI assistant to generate informative, context-aware, and personalized recommendations.

The system consists of four key agents:

Workout Recommendation Agent generates personalized home workout routines based on user inputs such as fitness goals, experience level, available time, and preferences.

Nutrition Agent suggests balanced meal ideas, healthy food options, and basic nutritional guidance according to dietary preferences and lifestyle.

Fitness Assistant Agent handles general fitness and wellness queries, providing interactive and context-aware responses through conversational AI.

Motivation Agent provides daily motivational tips, fitness inspiration, and habit-building suggestions to encourage users to maintain consistency.

The solution includes a **user-friendly dashboard** that displays workout history, daily fitness goals, completed activities, and overall progress toward personal fitness objectives.

LangGraph is used to orchestrate multi-agent workflows, while **IBM watsonx.ai and Granite Models** power AI-driven reasoning, response generation, and personalization.

The system also incorporates input and output guardrails to validate user queries, filter inappropriate content, and improve the safety and reliability of AI-generated responses.

Overall, this solution delivers an **end-to-end intelligent fitness companion** that promotes healthier lifestyles through personalized workout recommendations, nutrition guidance, continuous motivation, and progress tracking.

Problem Statement -

Problem Statement : Fitness Buddy

The Challenge
Maintaining a healthy lifestyle has become increasingly difficult due to busy schedules, lack of personalized fitness guidance, limited nutritional knowledge, and inconsistent motivation. Many individuals struggle with choosing suitable workouts, planning balanced meals, tracking their daily fitness habits, and maintaining consistency. Existing fitness solutions often require expensive subscriptions, personal trainers, or predefined routines that may not match individual needs and daily schedules. Without accessible and personalized guidance, people find it difficult to achieve their fitness goals and develop sustainable healthy habits.

The Objective
Build an **AI-powered Fitness Buddy** that provides personalized fitness and wellness guidance through an interactive conversational assistant. The solution should:
Personalized Workout Plans – Recommend home workouts based on fitness goals, experience levels, available time, and individual preferences.
AI-Powered Fitness Assistance – Answer fitness-related queries and provide context-aware recommendations through an interactive chatbot.
Nutrition & Meal Recommendations – Suggest simple, balanced, and nutritious meals based on dietary preferences and lifestyle.
Fitness Goal Tracking – Help users set daily fitness goals and monitor their workout and wellness progress.
Motivation & Habit Building – Provide daily motivational messages, fitness tips, and reminders to encourage consistency.
Multi-Agent System –
Workout Recommendation Agent (generates personalized workout routines)
Nutrition Agent (suggests healthy meals and basic nutritional guidance)
Fitness Assistant Agent (handles general fitness-related queries)
Motivation Agent (provides motivational tips and habit-building suggestions)
Progress Visualization Dashboard – Display workout history, daily activity, completed goals, and overall fitness progress.

Technology Used

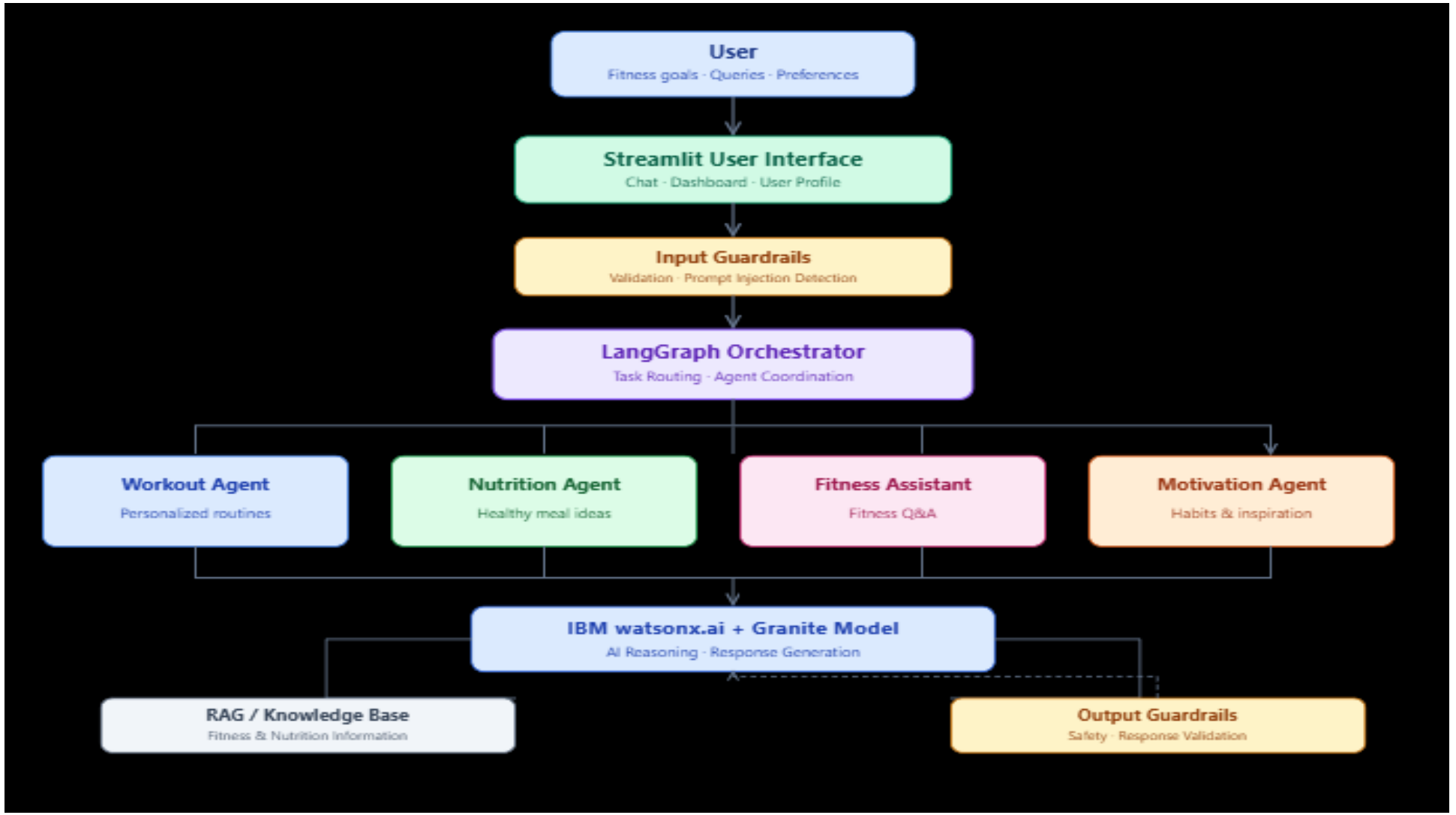
Langflow → LangGraph – For designing and orchestrating multi-agent workflows.
IBM Granite Model – ibm/granite-3-8b-instruct – For natural language understanding, reasoning, and personalized fitness recommendations.
IBM Cloud – Provides infrastructure for deploying and running the AI-powered Fitness Buddy application.
RAG (Retrieval-Augmented Generation) – Retrieves relevant fitness and nutrition information from trusted sources before generating informed responses.
IBM watsonx.ai – For Granite model integration, AI inference, and model deployment.
Vector Database – FAISS – For storing and retrieving relevant fitness and nutrition knowledge for RAG-based responses.
Python – For backend development, AI agent implementation, and application logic.
Streamlit – For building the interactive user interface and fitness dashboard.
Guardrails – For input validation, prompt injection detection, and safe AI-generated responses.

Langflow component Used -

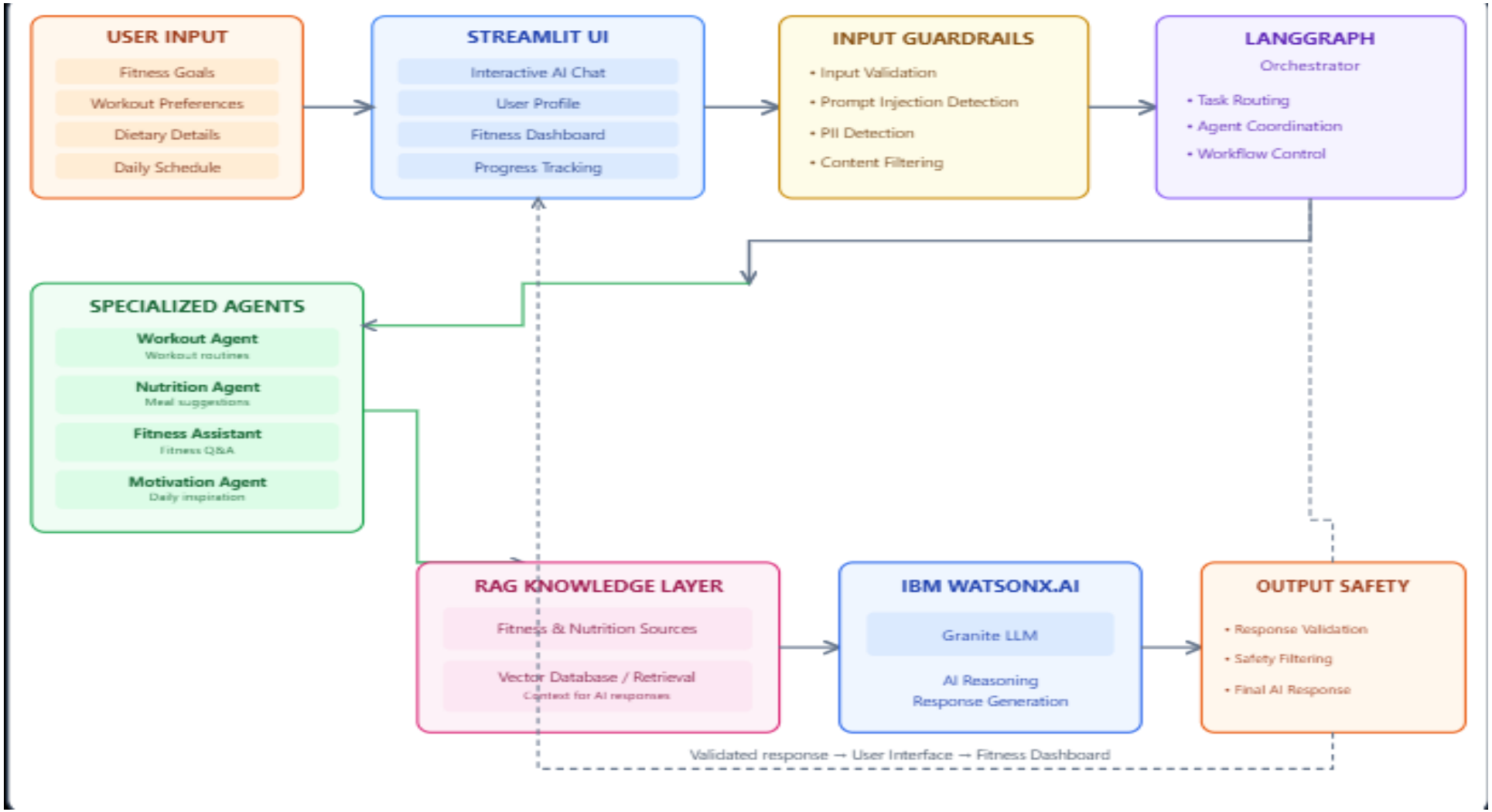
1. **Heading**: Langflow component Used → LangGraph Components Used
2. **Chat Input** – Users enter fitness queries, workout preferences, fitness goals, and dietary details.
3. **IBM watsonx AI Agent** – Processes user queries using IBM Granite models to generate personalized workout, nutrition, and wellness recommendations.
4. **Name of IBM Granite Model** – ibm/granite-3-8b-instruct
5. **Chat Output** – Displays AI-generated workout plans, healthy meal suggestions, fitness guidance, and motivational responses.
6. **File Component** – Stores and retrieves user fitness profiles, workout history, meal logs, and relevant fitness datasets.
7. **Add these other components:**

Workout Recommendation Agent – Generates personalized workout routines.
Nutrition Agent – Suggests healthy meals and nutritional guidance.
Fitness Assistant Agent – Handles fitness and wellness queries.
Motivation Agent – Provides daily motivation and habit-building suggestions.
Guardrails – Validates inputs and filters unsafe or inappropriate AI responses.
Dashboard – Visualizes fitness activities, completed goals, and progress.

Langflow workflow -



Architecture blueprint



Role of Agentic AI in the solution

Role of Agentic AI in Nutrition Agent

Agentic AI enables Fitness Buddy to function as an intelligent, goal-oriented virtual fitness assistant rather than a simple chatbot. Using **IBM watsonx.ai and IBM Granite models**, it coordinates multiple specialized agents to provide personalized fitness and wellness guidance.

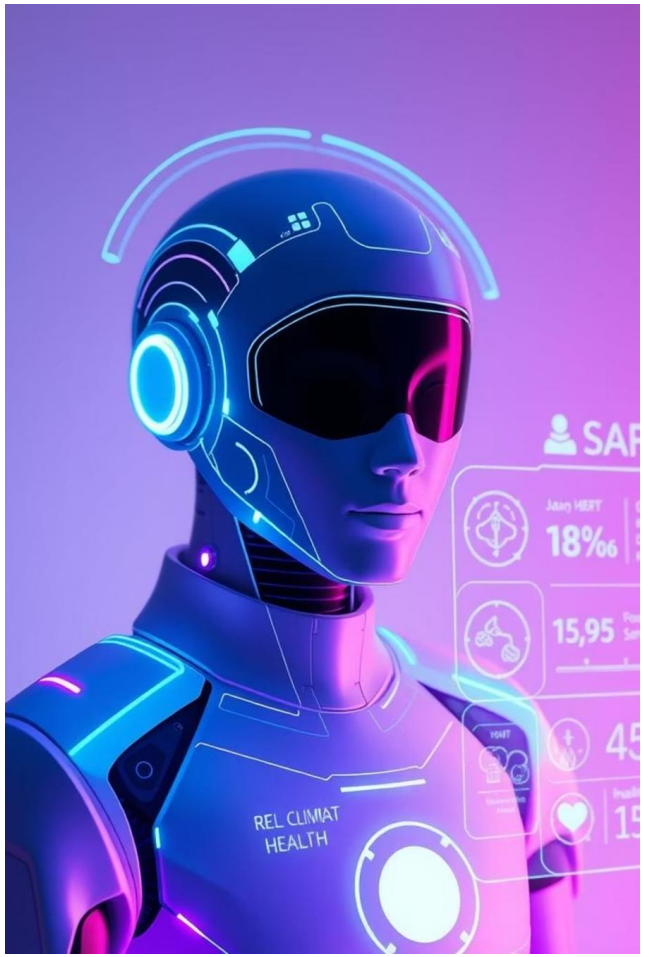
Each agent performs a specific role such as generating workout routines, suggesting healthy meals, answering fitness-related queries, and providing motivational guidance while working collaboratively. This **multi-agent approach** enables efficient task execution and personalized recommendations.

Agentic AI also brings **context awareness**, understanding user details such as fitness goals, experience levels, dietary preferences, available time, and daily routines to generate relevant suggestions.

It supports **adaptive recommendations**, modifying workout and wellness suggestions based on user preferences and feedback. Additionally, agents can collaborate to provide more comprehensive fitness guidance and help users make informed lifestyle decisions.

Through **automated task coordination and intelligent decision-making**, Agentic AI makes Fitness Buddy more interactive, personalized, and goal-driven.

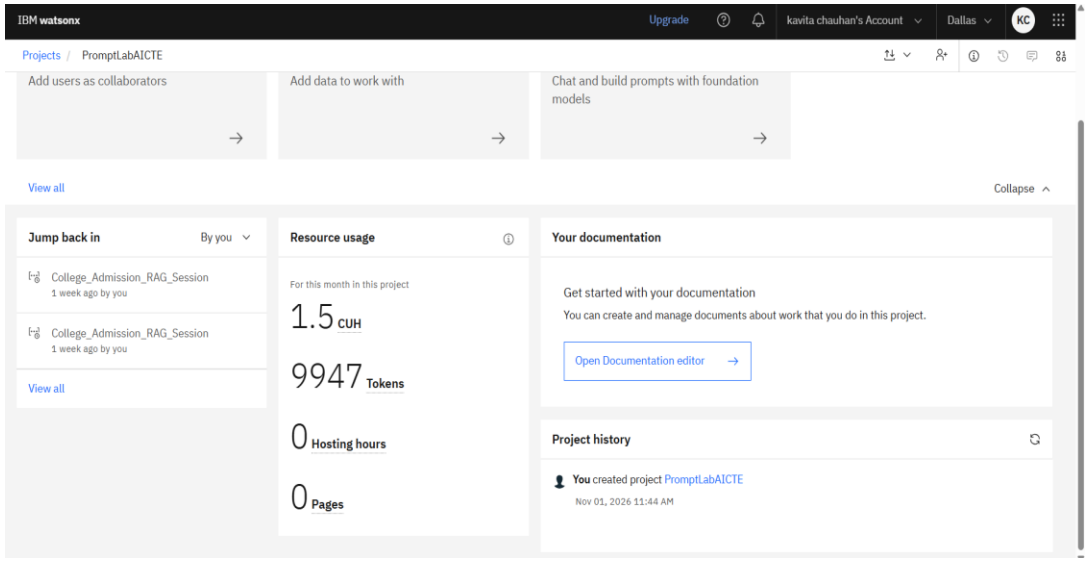
Overall, Agentic AI transforms Fitness Buddy into a **smart digital fitness companion** that helps users maintain consistency, achieve fitness goals, and develop healthier lifestyle habits.



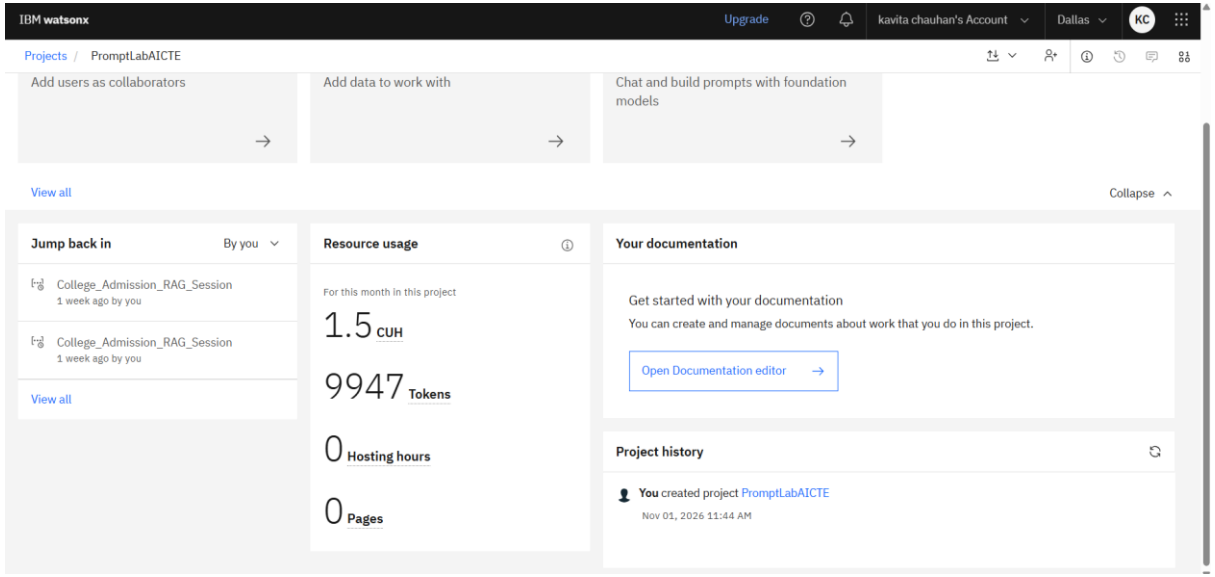
Token Usage

11,761 token are used

Before Prompt



After the Prompt



Novelty and Uniqueness

The proposed **Fitness Buddy** stands out by combining **Agentic AI, RAG, and personalization** into a single intelligent system.

- **Multi-Agent Intelligence** – Instead of a single chatbot, multiple specialized agents (Workout Agent, Nutrition Agent, Fitness Assistant, and Motivation Agent) collaborate to deliver accurate and context-aware fitness guidance.
- **RAG-Based Accuracy** – Integrates real-time retrieval of reliable fitness and nutrition information with AI generation, reducing misinformation and ensuring trustworthy recommendations.
- **Hyper-Personalization** – Provides tailored workout plans and meal suggestions based on fitness goals, health conditions, dietary preferences, experience level, and available time.
- **Real-Time Feedback Loop** – Continuously adapts recommendations based on user activity, workout history, meal logs, and user feedback.
- **Multi-Modal Input** – Supports text, voice, and image-based inputs (e.g., meal photos) for easier and more natural interaction.
- **Holistic Wellness Focus** – Goes beyond workouts by including nutrition guidance, habit building, and daily motivation to support overall wellness.
- **Scalable & Enterprise-Ready** – Built on **IBM watsonx.ai** with **IBM Granite models**, ensuring reliability, scalability, and AI governance.

This makes the solution a **smart, adaptive, and proactive digital fitness companion**, not just a static workout or diet recommendation tool.

Git Hub Link

Please create public github repository (Enable Add README button while creating repository)

with format –

Please test and Paste the working Github URL –

Make sure you have Uploaded following three files into your github repository

https://github.com/Kavita-Codes/AICTE-Fitness-Buddy

.

.

Future Scope

1. Integration with Wearables & Fitness Devices –
The system can be integrated with smartwatches and fitness trackers to collect real-time data such as steps, calories burned, heart rate, sleep patterns, and physical activity. This will enable Fitness Buddy to provide more accurate and personalized workout recommendations based on actual user activity and fitness progress.

2. AI-Powered Voice Assistant & Multilingual Support –
Enhancing the system with a voice-enabled assistant and support for multiple languages will improve accessibility. Users will be able to interact naturally in their preferred language, making Fitness Buddy more inclusive and convenient for people from diverse backgrounds.

3. Personalized Goal Tracking & Smart Notifications –
The system can introduce automated reminders, daily challenges, and personalized notifications to encourage consistency, improve motivation, and help users achieve their fitness goals.

Certificates Earned:



Certificates Earned:

IBM **SkillsBuild** Completion Certificate

This certificate is presented to

kavita chauhan

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Thank You!

Thank you for your time and interest.